

Interdisciplinary-driven generational shifts in vector mosquito surveillance technologies: from traditional traps to smart systems

Xiang Guo^{a,b,c}, Ziyao Li^b, Shu Zeng^b, Yun Zhou^a, Tieshan Teng^a, Yongzhi Li^d,
Yaohui Wang^{a,*}, Xiaohong Zhou^{b,**}, Xiao-Guang Chen^{b,***}

^a School of Basic Medical Sciences, Henan University, Kaifeng, 475004, China

^b Department of Pathogen Biology, Institute of Tropical Medicine, School of Public Health, Southern Medical University, Guangdong Provincial Key Laboratory of Tropical Disease Research, Key Laboratory of Prevention and Control for Emerging Infectious Diseases of Guangdong Higher Institutes, Key Laboratory of Infectious Diseases Research in South China of Ministry of Education, Guangzhou, 510515, China

^c Shenzhen Research Institute, Henan University, Shenzhen, 518000, China

^d School of Public Health, Zhaoqing Medical College, Zhaoqing, 526020, China

ARTICLE INFO

Keywords:

Mosquito surveillance
Generation system
AI
Mosquito-borne disease

ABSTRACT

Mosquito-borne infectious diseases, such as dengue fever and Chikungunya, are experiencing rapid and sustained increases in both geographic prevalence and epidemic scale, posing a significant and growing threat to global public health systems. From manual specimen collection to the development of traps in 1928 and the recent emergence of automated monitoring tools, interdisciplinary collaboration has served as the central driver of technological advancement. In the current era, propelled by innovations in perception technology, artificial intelligence (AI), Internet of Things (IoTs), and drones, mosquito surveillance is transitioning from traditional approaches toward intelligent, integrated systems. This evolution holds great promise for establishing efficient, reliable, and precise surveillance networks, thereby advancing the sustainable control of mosquito-borne diseases on a global scale.

1. Introduction

Mosquito-borne diseases have emerged as a critical global health challenge^{1–3}, with dengue fever alone causing an estimated 390 million infections annually.⁴ The geographical expansion of key vector species such as *Aedes albopictus*⁵ and *Anopheles stephensi*,⁶ driven by climate change and urbanization,^{7,8} has rendered traditional surveillance methods increasingly inadequate. While historical approaches like human landing catches and Breteau index (BI) provided foundational data for 20th-century control programs,^{9,10} their dependence on manual labor, subjective interpretation, and localized sampling limits their ability to capture the spatiotemporal dynamics of contemporary outbreaks.

The past decade has witnessed transformative innovations at the intersection of entomology and engineering, where advances in miniaturized sensors, machine learning, and networked devices are redefining

surveillance paradigms. For example, such automated traps, exemplified by Internet of Things (IoTs)-enabled platforms such as the MS300¹¹, transmit data to cloud-based dashboards, allowing public health authorities to monitor vector populations with unprecedented granularity.¹²

This review provides a systematic definition of generational transitions in mosquito surveillance technology and emphasizes the interplay between technological readiness and epidemiological utility across larval and adult monitoring pathways. Spanning 120 years of innovation, from manual aspirators to automated counter traps, we demonstrate how interdisciplinary advances have alternately served as key drivers of technological progress. We identify urgent needs for next-generation surveillance systems to advance cost-efficient and climate-resilient mosquito monitoring ecosystems, aligning with the priorities of the World Health Organization's (WHO) Global Vector Control Response.¹³

Peer review under the responsibility of Jiangsu Institute of Parasitic Diseases.

* Corresponding author.

** Corresponding author.

*** Corresponding author.

E-mail addresses: wangyaohui2017@henu.edu.cn (Y. Wang), daizhouxh@163.com (X. Zhou), xgchen@smu.edu.cn (X.-G. Chen).

<https://doi.org/10.1016/j.dcit.2026.100091>

Received 14 May 2026; Received in revised form 3 July 2026; Accepted 6 September 2026

Available online 12 September 2026

2949-9240/© 2026 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co., Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

2. Generation classification of current mosquito surveillance technology

We propose a generation-based classification system for current mosquito surveillance technologies, which is grounded in four key criteria: the level of automation and intelligence, operational convenience, objectivity versus subjectivity, and the degree of systematization (Fig. 1). The classification is structured along two primary technical pathways: monitoring of mature adult mosquitoes and immature larvae. Within the adult mosquito surveillance pathway, the first-generation technology encompasses manual methods such as the handheld mosquito suction device (Fig. 1). The second-generation technology involves the use of sensory attractants - such as sound, light, electricity, and chemical cues - to lure adult mosquitoes for capture (Fig. 1). The third-generation technology builds upon second-generation traps by integrating IoTs systems to enable automated counting of captured mosquitoes (Fig. 1). For larval surveillance, the first-generation approach includes manual inspections of breeding sites using BI methodology (Fig. 1). The second-generation technology employs artificial oviposition sites to attract gravid females for egg-laying, commonly referred to as ovitraps (Fig. 1). The third-generation technology enhances ovitraps with IoT and AI-based image recognition systems to enable automated detection and quantification of eggs or larvae (Fig. 1). This classification system provides a structured framework for evaluating the evolution and effectiveness of mosquito surveillance tools across developmental stages.

3. The first generation technology: classic and still not outdated

3.1. Human/animal-bait catches

Mosquito density monitoring initially utilized direct collection methods, such as mosquito nets or electric mosquito aspirators, in which humans or animals (e.g., pigs, mice, and rabbits) were employed as bait to attract and capture mosquitoes.^{14,15} Different mosquito species exhibit distinct diurnal biting patterns, leading to variations in species diversity and composition at different times of the day. For example, female *Ae. albopictus* were shown to bite frequently all day long, with bimodality occurring within two to 3 h of both dawn and dusk.^{16,17} Moreover, biting behaviors vary across geographic regions. *An. nunez-tovari* displays crepuscular host-seeking activity in many parts of the Amazon Region¹⁸, yet exhibits a nocturnal biting pattern in western

Venezuela and Colombia.^{18,19} Even at different altitudes, trapping often yields markedly different results. The location of the trapping site also significantly influences the outcomes. The variations in environmental factors, such as the spatial and temporal heterogeneity, can be effectively mitigated through the optimization of sampling techniques. Additionally, the influence of the capturer represents a critical factor that should not be overlooked in determining the outcomes.

However, this direct collection method inherently carries the risk of mosquito bites and potential disease transmission. To address this limitation, bed nets have been introduced as an improved protective measure.²⁰ The individual acting as bait can be enclosed within a fully protective mosquito net, effectively preventing direct contact with mosquitoes. Currently, the most widely adopted design is the human double-bait net (HDN),^{21,22} which comprises two nested box-shaped nets - one inner net that shields the human bait and a larger outer net placed directly above it. The outer net is elevated above the ground, creating a collection space between the two nets where mosquitoes attracted to the human bait can be captured. Various types of bait-protected trapping nets have been employed not only to capture mosquitoes attracted to humans, but also those that feed on a range of mammals and birds.¹⁴ Almost all openings and entrances of traps, including mosquito nets, present challenges for mosquitoes, which must overcome these barriers to enter and be captured. Consequently, fewer mosquito species are typically collected inside mosquito nets compared to those captured outside.²³ However, in a small subset of mosquito species, host movement plays a significant role in host detection. These species are therefore less likely to be effectively sampled when the host is protected by a mosquito net, as they tend to locate the host from external positions.

3.2. Larval indices

From the initial introduction of two larval indices—the House (or Premises) Index (HI) and the Container Index (CI)—in 1923²⁴, to the ongoing development of new indices in recent years, larval indices have remained a cornerstone of mosquito vector surveillance (Table 1). HI measures the percentage of houses infested with larvae and/or pupae, while CI reflects the proportion of water-holding containers harboring active immature stages of mosquitoes (Table 1). Subsequently, a series of newly proposed indices have largely followed the conceptual framework established by early larval indices (Table 1). For example, indices similar to the HI measure the number of breeding sites or larvae detected

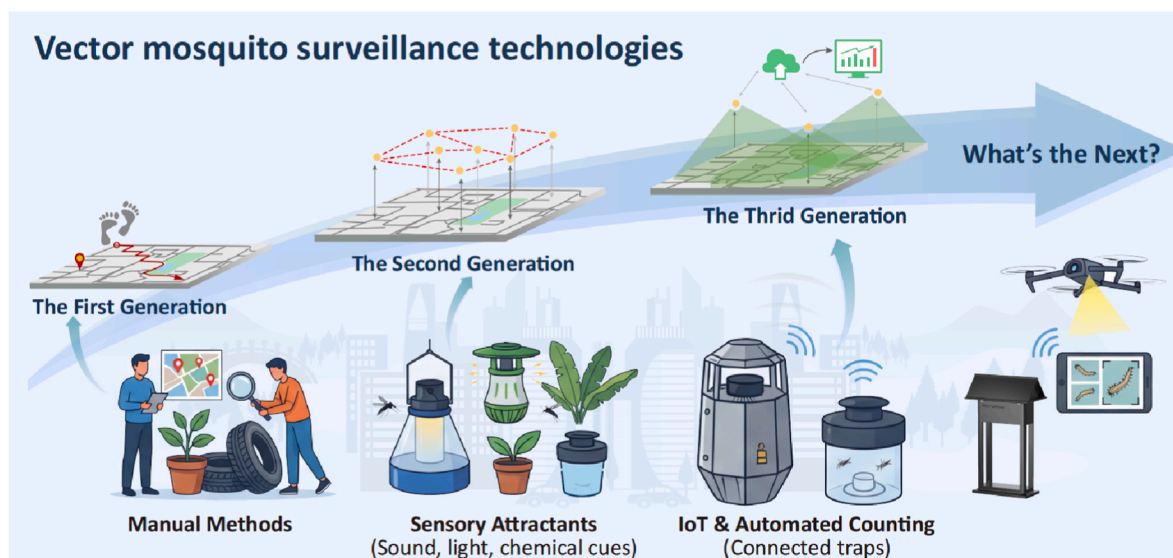


Fig. 1. Concept of current mosquito surveillance technology.

Table 1
Summary larval indices of vector mosquito surveillance.

Index	Abbr.	Year	Calculation	Reference
House index	HI	1923	$HI = \frac{\text{No. of houses positive for larvae}}{\text{No. of houses inspected}} \times 100$	24
Container index	CI	1923	$CI = \frac{\text{No. of positive containers}}{\text{No. of containers inspected}} \times 100$	24
Breteau index	BI	1954	$BI = \frac{\text{No. of Positive containers}}{\text{No. of house insepcted}} \times 100$	25
Block index	/	1967	Block index = the percent of the inspected blocks that are infested	33
Larval density index	LDI	1971	$LDI = \frac{\text{No. of larvae number}}{\text{No. of house insepcted}}$	26
Adult productivity index	API	1996	API = sum of container type $\times$ mean larvae	32
Pupal index	PI	2003	$PI = \frac{\text{No. of pupae number}}{\text{No. of house insepcted}} \times 100$	28
Standard space index	SSI	2006	$SSI = \frac{\text{No. of positive containers}}{\text{No. of standard room}} \times 100$	27
Route index	RI	2009	$RI = \frac{\text{No. of positive containers}}{\text{The number of kilometers walked}} \times 100$	9
Breeding Percentage	BP	2019	$BP = \frac{\text{No. of Ae. aegypti positive breeding sites}}{\text{No. of Aedes spp. positive breeding sites}} \times 100$	34

within a standardized survey area, including the BI,²⁵ Larval density index (LDI),²⁶ Standard space index (SSI),²⁷ Pupal index (PI),²⁸ and Route index (RI),⁹ among others (Table 1). The BI,²⁵ introduced in 1945, quantifies the number of positive containers per 100 houses and remains widely recommended as a practical technology in many regions (Table 1)^{9,29–31}. The Adult Productivity Index (API) represents another derivative of this approach.³² Similarly, indices aligned with the CI concept assess the proportion of positive containers within defined units, such as the Block Index.³³ A particularly notable index that diverges from these two conventional frameworks is the Breeding Percentage (BP), which was developed based on local mosquito species composition and dengue transmission dynamics, making it highly context-specific and operationally relevant.³⁴

4. The second generation technology: cost-effective solutions in current practice

4.1. Adult monitor for host-seeking, mating, and resting behavior

The concept of simulating host presence using physical stimuli like sound, light, and thermal energy, combined with fan-assisted capture mechanisms employing alternative baits, showed early promise. Evidence dates back to a 1928 practice in New Jersey that utilized a light trap exploiting the biological behavior of target species, primarily *Culex* and *Anopheles* mosquitoes.³⁵ Subsequent generations of adult mosquito traps saw limited technological advancement in their fundamental operating principles, which generally consist of three components: (1) an attractant; (2) fans and airflow-driven capture channels; and (3) a collection chamber or bag. Commercially available traps such as the classic Fay-Prince Trap,³⁶ BG-Trap,³⁷ Mosquito Magnet,³⁸ and Suna Trap³⁹ are all designed and deployed based on this foundational framework. As research into mosquito host-seeking behavior progressed, chemical receptors, particularly olfactory receptors, were increasingly identified, leading to the development and integration of various chemical attractants. For instance, the Fay-Prince Trap uses CO₂ as an attractant, significantly improving *Aedes* capture rates. In contrast, the Mosquito Magnet employs a multi-modal attraction strategy combining CO₂, Octenol Biting Insect Attractant, and thermal cues (Table 2).⁴⁰ Biogents developed the BG-Lure, a slow-release formulation that mimics human odour by releasing volatile compounds such as lactic acid, caproic acid, and ammonium bicarbonate, with optional supplementation of CO₂ in the BG-Trap system.³⁷

These low-cost, adult mosquito-specific monitoring devices developed in response to epidemiological needs are capable of flexibly adopting various integrated configurations, encompassing diverse

Table 2
Current traps belonging to the second generation technology of vector mosquito surveillance.

Behaviours	Year	Traps	Baits	Reference
Phototaxis	1928	CDC-light trap (CDC-LT)	Vision	35
Host seeking	1970	Fay-Prince Trap	Vision	36
	1979	Encephalitis Vector Survey trap (EVS Trap)	Odour	41
	1990s	Mosquito Magnet	Odour & vision	38
	2006	BG-Sentinel Trap (BGS)	Odour & vision	37
	2019	Suna trap	Odour	39,42
	2023	MTego	Odour	43
	2024	USDA-BHT	Vision, olfactotyr, and acousitc	44
Mating	2020	Male Aedes Sound Trap (MAST)	Sound	45,46
Resting	/	Resting trap	Thalposis & vision	47,48
Oviposition	1966	Oviposition trap (OT)	Vision	49
	1977	Autocidal gravid ovitrap (AGO)	Vision	50
	2003	Sticky ovitrap	Vision	51
	2005	Mosq-Ovitrap (MOT)	Vision	52
	2011	CDC-AGO trap	Vision	53
	2014	BG-GAT	Vision	54
	2025	Improved Mosquito Ovitrap (IMT)	Vision	55

physical designs and multiple bait combinations. For instance, the BG-Pro introduced by Biogents is a modular monitor that allows seamless switching between EVS-style, CDC-style, and Sentinel-style configurations (Table 2).⁵⁶ Different combinations of trapping methods exhibit varying levels of efficiency and suitability across different application environments (e.g., urban or rural areas) and among vector mosquito species.

The costs associated with maintenance, rather than production, have, on one hand, constrained the application of monitors targeting adult mosquitoes. Apart from light-trapping methods, attractants generally require frequent replacement within a relatively short time frame, and a reliable electrical power supply is essential for deploying such devices. The use of CO₂, which must be generated through various methods, such as dry ice, compressed propane catalytic canisters, yeast fermentation, or sodium bicarbonate combined with acid—significantly influences the accessibility, maintenance costs, and practical applicability of these monitors in underdeveloped regions, which notably bear a greater burden of mosquito-borne infectious diseases.

4.2. Gravid mosquitoes monitoring

Monitoring devices designed for gravid mosquitoes typically simulate optimal artificial breeding sites, thereby eliminating the need for an electrical power supply and enabling flexible deployment, which enhances their suitability for long-term outdoor surveillance. In 1965, Fay and Perry investigated the oviposition site preferences of female adult mosquitoes.⁵⁷ Subsequently, in 1966, Fay and Eliason developed the first Oviposition Trap (OT) (Table 2).⁴⁹ Since then, all ovitraps designed for gravid mosquitoes have been variations of this original design, incorporating the following key features: (1) a dark-colored container (often with high-contrast black and white surfaces); (2) a liquid medium (commonly plain water, plant extract infusions, oviposition attractants, or larvicides); (3) a substrate provided for egg-laying (sometimes treated to be sticky for capturing gravid females); (4) small, escape-proof openings or a labyrinth-style lid to prevent mosquito escape.

The ovitrap is known for its exceptionally high sensitivity in that it can still successfully collect eggs and even gravid adult mosquitoes when adult mosquito traps yield zero catches, implying the potential presence of virus-carrying vectors during epidemics.⁵⁸ The container size is closely related to the efficacy of gravid mosquito monitors, with larger containers and greater water volumes exhibiting stronger attraction to gravid mosquitoes.⁵⁹ For instance, the 9.2 L BG-GAT, on average, 3.7 more female *Ae. aegypti* and 2.4 more female mosquitoes in total compared to the 1.2 L BG-GAT.⁵⁹ Incidentally, if the water contains larvicidal agents or the oviposition surface is sticky, these monitors can also eliminate the mosquitoes population.^{51,60,61}

In practical applications, a balance must be carefully maintained among the monitor's volume, the number of deployed monitors, and the operational workload of personnel. Larger monitoring volumes and increased sensitivity can reduce the number of required monitors.⁵⁵ As monitoring expands from natural breeding sites to artificial ones, the evaluation index is correspondingly extended, primarily calculated as the ratio of positive containers to the total number of inspected containers to estimate vector mosquito density. A limitation in current practice is that the actual number of eggs collected by each monitor is often disregarded and aggregated solely under the binary classification of "positive containers" in index computation (for example, OI⁶² and MOI^{52,63}). In reality, variations in egg counts reflect differing levels of gravid mosquito activity and population density. The New Ovitrap Index (NOI) addresses this limitation by introducing an egg quantity-based risk assessment framework, which has been preliminarily implemented during outbreak responses.⁵⁵ Although it should be acknowledged that the shortened monitoring period and the requirement for precise egg counting have moderately increased the workload for field personnel.

5. The third generation technology: large-scale deployment in progress

Automation represents the defining characteristic of third-generation mosquito surveillance technology. Current limit systems such as MS300¹¹, BG-Counter,^{64,65} and VECTRACK^{66,67} are built upon trap designs analogous to the BGS, employing chemical lures in conjunction with fans to capture mosquitoes (Table 3). As mosquitoes

Table 3
Current traps belonging to the third generation technology of vector mosquito surveillance.

Machine	Year	Sensor	Reference
Sun, et al	2021	Vision camera	68
MS300	2022	Infrared sensor	11
Liu, et al	2023	Vision camera	69
Bg-Counter	2024	Infrared sensor	64,65
VECTRACK	2024	Infrared sensor	66,67
Maria, et al	2024	Optical wingbeat sensor	70,71

are drawn into the device, integrated sensors embedded within the airflow pathway enable real-time detection. These data are then transmitted via IoT connectivity to cloud-based platforms for automated analysis, monitoring, and actionable insights.

Despite the promise of third-generation and next-generation tools, critical evaluations of their field performance, validation standards, and practical limitations remain scarce. Currently, IoT-integrated surveillance systems including the MS300, BG-Counter, and VECTRACK have not achieved large-scale adoption or been embedded into routine workflows of any national mosquito surveillance program. Their real-world application is still predominantly restricted to academic research initiatives and geographically limited small-scale pilot projects. On one hand, the field trapping efficacy and overall species identification sensitivity of currently available devices still require further targeted optimization. On the other hand, the large-scale rollout of a connected national surveillance network faces constraints from prohibitively high upfront costs that demand substantial long-term investment. This makes a commercially sustainable, scalable business model an essential prerequisite to drive mass production and progressive cost reduction for widespread accessibility. Furthermore, compared with the well-standardized first-generation manual surveillance methods, existing IoT-integrated trap systems are still in urgent need of a unified, well-defined monitoring index system to support consistent, reliable vector density measurement, which severely hinders their adoption in routine public health decision-making.

The development of advanced technologies and instruments involves substantial interdisciplinary complexity. As exemplified by the MS300 system (Fig. 2), its full design lifecycle demands integrated expertise across multiple domains: life sciences including molecular biology, entomology and ecology to decode mosquito behaviors such as host-seeking; engineering disciplines covering automation, fluid dynamics, sensor technology and artificial intelligence (AI) for hardware and algorithm optimization; technical capabilities for natural plant-derived attractant synthesis, lure formulation, IoT transmission, full-stack software development and standardized data storage; as well as epidemiology, biostatistics and public health management to establish validated reference indices and frame compliant data governance. These interconnected fields follow the "barrel effect" principle, where the weakest link in the interdisciplinary chain acts as a critical bottleneck that constrains overall system performance. Ongoing large-scale deployment of the MS300 requires expanded, coordinated cross-disciplinary collaboration to address remaining technical and operational challenges.

6. The next generation tools: smart systems

An analysis of historical trends in mosquito vector monitoring technologies reveals that cross-disciplinary innovation has consistently driven technological advancement. The widespread adoption of second-generation systems aligned with breakthroughs in plastic manufacturing, which enabled cost-effective and scalable trap designs. In contrast, third-generation technologies emerged in parallel with the development of IoT and sensor networks, allowing for real-time data collection and remote monitoring capabilities. Looking ahead, the integration of biotechnology (BT), artificial intelligence (AI), and IoT provides a robust foundation for next-generation vector surveillance systems. Emerging research highlights promising developments in drone-based surveillance, AI-powered image recognition for species identification, computer vision, acoustic monitoring, radar, and Light Detection and Ranging (LiDAR)-assisted spatial mapping of mosquito habitats, collectively indicating a transformative trajectory in monitoring capabilities. These advancements mark a shift from the passive lure-and-capture mechanisms of earlier devices toward intelligent, precision-driven identification.

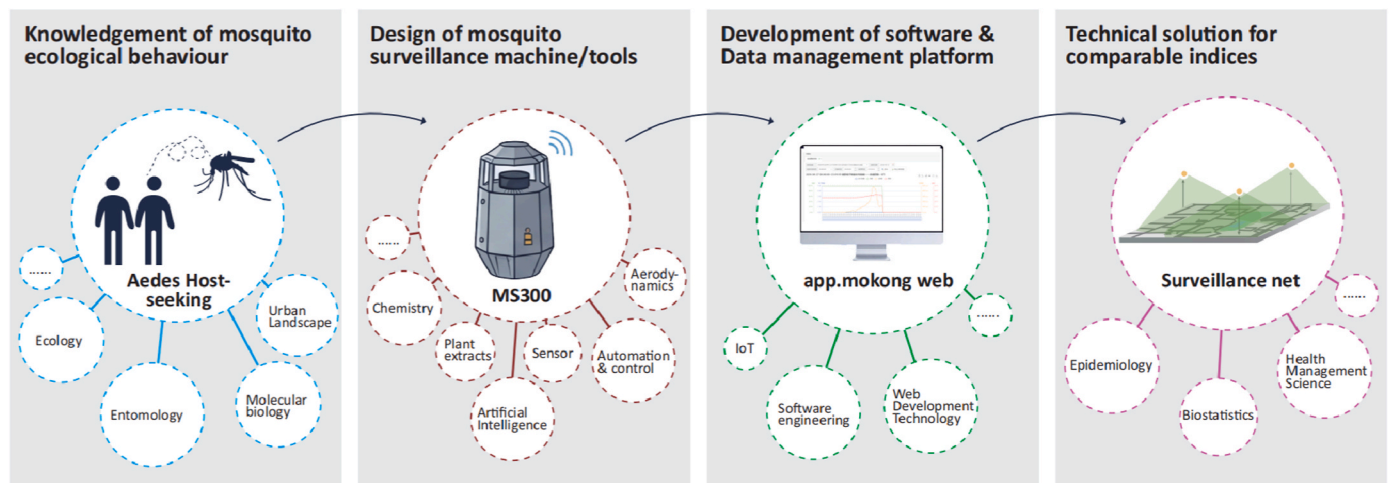


Fig. 2. Multidisciplinary integration for advanced mosquito surveillance technologies.

6.1. Multimodal fusion of perception technology

Perception technologies, including computer vision, acoustic monitoring, and LiDAR, have undergone significant advancements in recent years, enabling widespread deployment in autonomous vehicles, unmanned aerial systems, and consumer electronics such as smartphones. These technologies are now being actively explored for application in insect monitoring, marking an emerging frontier in ecological surveillance. Within the realm of ecology, computer vision is increasingly employed to automate the collection of observational data and facilitate species identification. For example, researchers often position cameras to capture specific environmental features or deploy imaging systems in natural habitats, frequently integrating various trapping methods to improve detection efficiency. Current approaches predominantly rely on machine learning algorithms—particularly convolutional neural networks (CNNs)—to identify insects by training on pre-labeled image databases. However, several critical technical challenges should be considered. Firstly, mosquitoes, being typically small-bodied, necessitate high-resolution imaging systems and precise focus tracking for accurate detection. Secondly, CNNs require large-scale, high-quality training datasets (i.e., reference image libraries), which must often be tailored to specific taxonomic groups and sensor configurations. Consequently, while the current AI-based species identification approach cannot train and recognize all mosquito species comprehensively, it remains effective for key vector mosquitoes in urban environments, such as *Ae. albopictus*. For these species, sufficient training data is readily available, enabling the development of robust identification models.

Acoustic monitoring employs field-deployed acoustic sensors to collect sound data and integrates machine-learning algorithms for species identification.⁷² Transect surveys using stationary acoustic sensors can record insect vocalizations.⁷³ To mitigate environmental noise interference, mosquitoes can be lured into specialized pipes or other controlled environments during data collection. But the capability of sound-based species identification is still limited by the size of existing reference databases.

Recent advances in remote sensing have enabled complementary approaches for aerial insect detection. Among these, LiDAR, an active optical technology that emits pulsed laser beams and analyzes their backscattered signals from airborne targets, has been successfully validated for long-range near-ground insect monitoring across a 10–600 m operational envelope.⁷² By quantifying core observables including time-of-flight, Doppler shift, and signal depolarization, LiDAR systems can simultaneously retrieve multi-dimensional traits of flying insects: geospatial distance, flight velocity, wing-beat frequency, and even 3D

body orientation, delivering spatially resolved, continuous datasets that map fine-scale mosquito activity dynamics and dispersal patterns.⁷² In contrast, Light-Emitting Diode (LED)-based optical sensors operate exclusively within a close-proximity detection range of ≤ 1 m, and are optimized for high-temporal-resolution tracking of individual insect specimens. Their inherently limited sampling volume, however, poses a fundamental constraint that prevents scalable, area-wide population-level surveillance.^{72,74} Looking ahead, iterative technological refinements paired with the systematic expansion of standardized spectral libraries covering diverse insect taxa are projected to enable robust, highly accurate automated species classification in the future.⁷²

Complementary to existing approaches, thermal imaging is an optional technology for mosquito monitoring, by detecting insects through their metabolic heat and activity-induced temperature changes. This is particularly applicable to larval surveillance in eutrophic waters, where organic matter generates characteristic “thermal filaments” or “hot spots”, contrasting with the homogeneous thermal signature of clear water.

6.2. Integrating drones into vector surveillance

A readily conceivable solution involves using drones equipped with high-resolution cameras to detect and identify vector breeding sites for surveillance purposes^{75–77}. For example, a study conducted in Thailand demonstrated the feasibility of breeding site identification using drone imagery, while further integrated recurrent neural networks (RNNs) to enhance the recognition of breeding sites.⁷⁶ From the standpoint of breeding site monitoring, index development could adopt the concept of RI, whereby drones scan target areas and quantify breeding sites per unit area to generate a standardized metric. However, the effectiveness of this approach is currently constrained by limitations in camera image resolution and the accuracy of automated detection models. Notably, *Ae. albopictus* commonly breeds in small often transient water bodies in urban environments,⁷⁸ which presents a significant challenge for drone-based imaging and scanning due to the low visibility and spatial scale of these habitats.

Another promising approach involves using drones equipped with insect nets for mosquito specimen collection.⁷⁹ The highest number of mosquitoes is captured at a height of 5.00 m, while collection efficiency drops significantly above 12.00 m or under wind speeds exceeding 4.00 m/s.⁷⁹ Although mosquitoes are predominantly distributed in lower altitudes, the downwash airflow generated by drone rotors can induce vertical movement of mosquitoes, causing them to fly upward or downward.⁷⁹ This disturbance may introduce discrepancies between observed collection patterns and their true spatial distribution,

highlighting the need for further research to clarify the impact of drone operation on sampling accuracy.

6.3. Can environmental DNA be a complementary?

Over the past decade, research has shown that environmental DNA (eDNA) sampling significantly improves invertebrate detection and provides insights into community composition.^{80,81} The most common approach is DNA metabarcoding, which can be applied directly to preserved specimens or pooled insect homogenates.⁸² By extracting and sequencing DNA from environmental samples—including water, soil, and air—species presence can be reliably detected. Nevertheless, despite its broad applicability, the use of eDNA for detecting mosquito species has remained underexplored.^{83,84}

A study was conducted in which water samples were collected from potential larval breeding sites along a human disturbance gradient extending from the center of an oil palm plantation to adjacent rainforest areas on São Tomé Island, Gulf of Guinea, Africa.⁸³ Metabarcoding analysis revealed that most mosquito species were detectable via eDNA when larvae were visually present; compared to traditional morphological identification, eDNA yielded higher larval species richness and additionally detected previously unrecorded mosquito species.⁸³ However, metabarcoding technology still faces several unresolved technical challenges. These challenges arise from multiple interrelated factors. In addition to the pronounced variation in DNA content and extraction efficiency across insect taxa, taxon-specific PCR biases can lead to false-negative results even when target species are present. Moreover, the primary outputs of metabarcoding, amplicon sequence variants (ASVs) and operational taxonomic units (OTUs), are highly dependent on the bioinformatic pipeline used. Consequently, assigning these molecular units to known species requires comparison against reference databases such as the Barcode of Life Data System (BOLD) or GenBank. However, limitations of these databases, including sequencing errors, synonymy, misidentifications, and incomplete taxonomic coverage, can result in erroneous taxonomic assignments. Beyond these general eDNA-related constraints, practical challenges such as sampling representativeness of breeding site water, cost-effectiveness, and the lack of standardized monitoring frameworks, including performance metrics, remain significant barriers to the widespread and reliable application of this technology in the near term.

7. Conclusion and outlook

Above all, the evolution of mosquito surveillance technologies, progressing from manual collection workflows to fully integrated AI-powered smart systems, vividly demonstrates the transformative capacity of interdisciplinary innovation to reshape public health strategies for vector-borne disease control. First-generation tools laid the foundational framework by establishing standardized core surveillance metrics, while second-generation traps delivered marked improvements in cost-effectiveness through targeted advances in material science. Modern third-generation systems have further unlocked the revolutionary potential of IoT connectivity and machine learning, enabling real-time, data-driven vector population management. As the field advances toward next-generation solutions that integrate drones, eDNA profiling and multi-modal sensing modules, it has become an urgent priority to systematically address critical unmet challenges: scalable cost reduction, unified data standardization, large-scale network deployment, and the development of intelligent decision-making platforms, to establish a fully closed-loop automated "monitoring-decision-intervention" system.

The generational classification of surveillance technologies outlined in this work not only provides a structured, interpretable framework for understanding the field's evolutionary trajectory, but also establishes a consensus foundation to shift the research focus toward a flexible, context-aware integrated technical paradigm. This paradigm leverages the unique inherent advantages of each generation of tools, to develop

efficient, high-performance surveillance strategies that support tiered deployment across complex environmental and socioeconomic settings, and satisfy the differentiated demands of both routine long-term monitoring and emergency outbreak response. It delivers novel actionable surveillance and control frameworks, robust technical support, and a clear forward-looking roadmap to curb the rapid global spread of dengue, Chikungunya and other mosquito-borne infectious diseases.

This paradigm shift, transitioning from reactive, labor-intensive surveillance to proactive, intelligent end-to-end ecosystems, promises to fundamentally enhance our capacity for pre-emptive outbreak mitigation, by integrating targeted surveillance of vector population density to move epidemic prevention controls upstream, paired with predictive analytics and automated intervention workflows. By consolidating advanced cross-domain technologies, we can build a more responsive, adaptive surveillance system that not only identifies potential public health threats at the earliest stage, but also deploys timely, precision-targeted control measures tailored to complex, dynamic prevention and control requirements. Delivering this full vision requires sustained, coordinated interdisciplinary collaboration, with all technological iterations consistently anchored to core epidemiological priorities and verified field applicability across resource-constrained and well-resourced settings alike. Only through such aligned cross-sector collaboration can we develop solutions that are both technically sophisticated, and practically feasible and cost-effective under real-world deployment conditions.

There is an urgent, unaddressed need to establish unified standardized performance metrics and formal validation protocols for automated and semi-automated mosquito surveillance devices. Currently, widely deployed systems including the BG-Counter, MS300 and VECTRACK are evaluated under highly heterogeneous experimental conditions (covering variable mosquito density gradients, trap placement configurations and climatic contexts), making direct, unbiased cross-device performance comparison extremely challenging. We explicitly recommend the global public health and entomology community adopt a minimum unified reporting framework covering four core dimensions: detection sensitivity, species identification specificity, validated field reliability, and operational maintenance burden. This framework will empower end-users to make evidence-based procurement and deployment decisions, while incentivizing manufacturers to iteratively improve device performance through transparent, verifiable iterations, and support the gradual establishment of a complete corresponding technical guideline and standard system. Furthermore, alongside the rollout of this standardized evaluation framework, a system-level coordinated effort is urgently required to harmonize instrument data interfaces across the full ecosystem, unifying hardware communication protocols, standardized data schemas and interoperable metadata specifications, and build robust, scalable data governance and management infrastructures that guarantee full interoperability, traceability and long-term usability across all heterogeneous surveillance platforms.

In essence, the future of mosquito surveillance hinges on the seamless integration of cumulative insights from each technological generation and context-adaptive deployment strategies, tailored to address the dynamic evolving challenges and diversified tiered prevention and control demands of public health amid continuously emerging vector-borne disease threats. By fully leveraging the complementary strengths of every generation of surveillance tools and fostering deep, sustained interdisciplinary cooperation, we can pave a clear path toward a more resilient, effective and cost-efficient public health response system for mosquito-borne disease control.

CRediT authorship contribution statement

Xiang Guo: Writing – original draft, Visualization, Funding acquisition, Conceptualization. **Ziyao Li:** Writing – original draft. **Shu Zeng:** Writing – original draft. **Yun Zhou:** Writing – review & editing. **Tieshan Teng:** Writing – review & editing. **Yongzhi Li:** Writing – review &

editing. **Yaohui Wang:** Writing – review & editing. **Xiaohong Zhou:** Writing – review & editing, Conceptualization. **Xiao-Guang Chen:** Funding acquisition, Conceptualization.

Ethics approval and consent to participate

Not applicable.

Availability of data and materials

Not applicable.

Funding

This study was supported by Guangdong S&T programme, China (2025B1111050002), the National Natural Science Foundation of China, China (82502757), and the 'Revealing the Leader' Program of the Guangdong Provincial Patriotic Health Office's Research and Application of Innovative Technologies for Disease Vector Monitoring and Control, China.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

None.

References

- Zhang M, Li Y, Huang X, et al. Epidemiological characteristics and transmission dynamics of the early stage Chikungunya fever outbreak in Foshan City, Guangdong Province, China in 2025. *Infect Dis Poverty*. Sep 11 2025;14(1):93. <https://doi.org/10.1186/s40249-025-01364-y>.
- Petersen LR, Jamieson DJ, Powers AM, Honein MA. Zika virus. *N Engl J Med*. Apr 21 2016;374(16):1552–1563. <https://doi.org/10.1056/NEJMr1602113>.
- Messina JP, Brady OJ, Golding N, et al. The current and future global distribution and population at risk of dengue. *Nat Microbiol*. Sep 2019;4(9):1508–1515. <https://doi.org/10.1038/s41564-019-0476-8>.
- Bhatt S, Gething PW, Brady OJ, et al. The global distribution and burden of dengue. *Nature*. Apr 25 2013;496(7446):504–507. <https://doi.org/10.1038/nature12060>.
- Kraemer MUG, Reiner Jr RC, Brady OJ, et al. Past and future spread of the arbovirus vectors *Aedes aegypti* and *Aedes albopictus*. *Nat Microbiol*. May 2019;4(5):854–863. <https://doi.org/10.1038/s41564-019-0376-y>.
- Ryan SJ, Lippi CA, Villena OC, Singh A, Muddock CC, Johnson LR. Mapping current and future thermal limits to suitability for malaria transmission by the invasive mosquito *Anopheles stephensi*. *Malar J*. Mar 21 2023;22(1):104. <https://doi.org/10.1186/s12936-023-04531-4>.
- de Souza WM, Weaver SC. Effects of climate change and human activities on vector-borne diseases. *Nat Rev Microbiol*. Aug 2024;22(8):476–491. <https://doi.org/10.1038/s41579-024-01026-0>.
- Erizzo D, Grant L, Ghisbain G, et al. Contribution of climate change to the spatial expansion of West Nile virus in Europe. *Nat Commun*. Feb 8 2024;15(1):1196. <https://doi.org/10.1038/s41467-024-45290-3>.
- Surveillance methods for vector density - mosquito. Beijing: General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China and Standardization Administration of the People's Republic of China. 2009.
- Liu X, Liu Q. Aedes surveillance and risk warnings for Dengue - china, 2016–2019. *China CDC Wkly*. Jun 12 2020;2(24):431–437. <https://doi.org/10.46234/ccdcw2020.111>.
- Lai Z, Wu J, Xiao X, et al. Development and evaluation of an efficient and real-time monitoring system for the vector mosquitoes, *Aedes albopictus* and *Culex quinquefasciatus*. *PLoS Neglected Trop Dis*. Sep 2022;16(9), e0010701. <https://doi.org/10.1371/journal.pntd.0010701>.
- He Q, Lu Q, Xie N, et al. Real-time dynamic monitoring and multiplex PCR identification of vector mosquitoes in Zhejiang, China. *PLoS Neglected Trop Dis*. May 2025;19(5), e0013129. <https://doi.org/10.1371/journal.pntd.0013129>.
- World Health Organization. Global vector control response: progress in planning and implementation. Accessed 2025/09/26/ <http://www.who.int/stable/resre/p35671>; 2020.
- Liu XB, Liu QY, Guo YH, et al. The abundance and host-seeking behavior of culicine species (Diptera: culicidae) and *Anopheles sinensis* in Yongcheng city, People's Republic of China. *Parasites Vectors*. Nov 24 2011;4:221. <https://doi.org/10.1186/1756-3305-4-221>.
- Darbro JM, Harrington LC. Bird-baited traps for surveillance of West Nile mosquito vectors: effect of bird species, trap height, and mosquito escape rates. *J Med Entomol*. Jan 2006;43(1):83–92. <https://doi.org/10.1093/jmedent/43.1.83>.
- Yin Q, Li L, Guo X, et al. A field-based modeling study on ecological characterization of hourly host-seeking behavior and its associated climatic variables in *Aedes albopictus*. *Parasites Vectors*. Oct 14 2019;12(1):474. <https://doi.org/10.1186/s13071-019-3715-1>.
- Guo X, Li Z, Zhao S, et al. Insights into the autecology of *Aedes albopictus*. *Parasites Vectors*. Nov 4 2025;18(1):445. <https://doi.org/10.1186/s13071-025-07032-2>.
- Rubio-Palis Y, Curtis CF. Biting and resting behaviour of anophelines in western Venezuela and implications for control of malaria transmission. *Med Vet Entomol*. Oct 1992;6(4):325–334. <https://doi.org/10.1111/j.1365-2915.1992.tb00628.x>.
- Elliott R. Studies on man vector contact in some malarious areas in Colombia. *Bull World Health Organ*. 1968;38(2):239–253.
- Colless DH. Notes on the culicine mosquitoes of Singapore. VI. Observations on catches made with baited and unbaited trap-nets. *Ann Trop Med Parasitol*. Sep 1959;53:251–258. <https://doi.org/10.1080/00034983.1959.11685922>.
- Tangena JA, Thammavong P, Hiscox A, Lindsay SW, Brey PT. The human-baited double net trap: an alternative to human landing catches for collecting outdoor biting mosquitoes in Lao PDR. *PLoS One*. 2015;10(9), e0138735. <https://doi.org/10.1371/journal.pone.0138735>.
- Lima JB, Galardo AK, Bastos LS, Lima AW, Rosa-Freitas MG. MosqTent: an individual portable protective double-chamber mosquito trap for anthropophilic mosquitoes. *PLoS Neglected Trop Dis*. Mar 2017;11(3), e0005245. <https://doi.org/10.1371/journal.pntd.0005245>.
- Gao Q, Wang F, Lv X, et al. Comparison of the human-baited double net trap with the human landing catch for *Aedes albopictus* monitoring in Shanghai, China. *Parasites Vectors*. Aug 28 2018;11(1):483. <https://doi.org/10.1186/s13071-018-3053-8>.
- Connor ME, Monroe WM. Stegomyia indices and their value in yellow fever control. *Am J Trop Med Hyg*. 1923;1(1):9–19. <https://doi.org/10.4269/ajtmh.1923.s1-3.9>.
- Breteau H. [Yellow fever in French West Africa; an aspect of mass-preventive medicine]. *Bull World Health Organ*. 1954;11(3):453–481. La fièvre jaune en Afrique-Occidentale Française; un aspect de la médecine préventive massive.
- Chan YC, Chan KL, Ho BC. *Aedes aegypti* (L.) and *Aedes albopictus* (Skuse) in Singapore City. 1. Distribution and density. *Bull World Health Organ*. 1971;44(5):617–627.
- Yan Z, Hu Z, Jiang Y, Wu H, Pan Z. Study on the application of standard space index in the population dynamic surveillance of *Aedes albopictus*. *Chin J Hygien Insect Equip*. 2006;12(5):352–354.
- Guidelines for Dengue Surveillance and Mosquito Control. World Health Organization; 2003.
- Aryaprema VS, Xue RD. Breteau index as a promising early warning signal for dengue fever outbreaks in the Colombo District, Sri Lanka. *Acta Trop*. Nov 2019;199, 105155. <https://doi.org/10.1016/j.actatropica.2019.105155>.
- Sanchez L, Vanlerberghe V, Alfonso L, et al. *Aedes aegypti* larval indices and risk for dengue epidemics. *Emerg Infect Dis*. May 2006;12(5):800–806. <https://doi.org/10.3201/eid1205.050866>.
- Morales-Perez A, Nava-Aguilera E, Hernandez-Alvarez C, et al. Utility of entomological indices for predicting transmission of dengue virus: secondary analysis of data from the Camino Verde trial in Mexico and Nicaragua. *PLoS Neglected Trop Dis*. Oct 2020;14(10), e0008768. <https://doi.org/10.1371/journal.pntd.0008768>.
- Tun-Lin W, Kay BH, Barnes A, Forsyth S. Critical examination of *Aedes aegypti* indices: correlations with abundance. *Am J Trop Med Hyg*. May 1996;54(5):543–547. <https://doi.org/10.4269/ajtmh.1996.54.543>.
- Tinker ME. Measurement of *Aedes aegypti* populations. *J Econ Entomol*. Jun 1967;60(3):634–637. <https://doi.org/10.1093/jee/60.3.634>.
- Ong J, Liu X, Rajarethinam J, Yap G, Ho D, Ng LC. A novel entomological index, *Aedes aegypti* Breeding Percentage, reveals the geographical spread of the dengue vector in Singapore and serves as a spatial risk indicator for dengue. *Parasites Vectors*. Jan 8 2019;12(1):17. <https://doi.org/10.1186/s13071-018-3281-y>.
- Mulhern T. A new development in mosquito traps. *Proceed Annual Meet New Jersey Mosquito Exterm Assoc*. 1934;20:137.
- Fay RW, Williams H. A modified visual trap for *Aedes aegypti*. *Mosq news*. 1970;30:20–23.
- Krockel U, Rose A, Eiras AE, Geier M. New tools for surveillance of adult yellow fever mosquitoes: comparison of trap catches with human landing rates in an urban environment. *J Am Mosq Control Assoc*. Jun 2006;22(2):229–238. [https://doi.org/10.2987/8756-971X\(2006\)22\[229:NTFSOA\]2.0.CO;2](https://doi.org/10.2987/8756-971X(2006)22[229:NTFSOA]2.0.CO;2).
- Kline DL. Comparison of two American biophysics mosquito traps: the professional and a new counterflow geometry trap. *J Am Mosq Control Assoc*. Sep 1999;15(3):276–282.
- Hiscox A, Otieno B, Kibet A, et al. Development and optimization of the Suna trap as a tool for mosquito monitoring and control. *Malar J*. Jul 7 2014;13:257. <https://doi.org/10.1186/1475-2875-13-257>.
- Garcia R, Colwell AE, Voigt WG, Woodward DL. Fay-Prince trap baited with CO2 for monitoring adult abundance of *Aedes sierrensis* (Diptera: Culicidae). *J Med Entomol*. Jul 1989;26(4):327–331. <https://doi.org/10.1093/jmedent/26.4.327>.
- Rohe D, Fall R. A miniature battery powered CO2 baited light trap for mosquito borne encephalitis surveillance. *Bull Soc Vector Ecol*. 1979;4:24–27.
- Verhulst NO, Bakker JW, Hiscox A. Modification of the Suna Trap for improved survival and quality of mosquitoes in support of epidemiological studies. *J Am Mosq*

- Control Assoc. Sep 2015;31(3):223–232. <https://doi.org/10.2987/moco-31-03-223-232.1>.
43. Maasayi MS, Machange JJ, Kamande DS, et al. The MTego trap: a potential tool for monitoring malaria and arbovirus vectors. *Parasites Vectors*. Jun 27 2023;16(1):212. <https://doi.org/10.1186/s13071-023-05835-9>.
 44. Lado P, Rogers DC, Cernicchiaro N, et al. Assessment of the USDA Biomass Harvest Trap (USDA-BHT) device as an insect harvest and mosquito surveillance tool. *J Econ Entomol*. Aug 12 2024;117(4):1235–1241. <https://doi.org/10.1093/jee/toae095>.
 45. Staunton KM, Crawford JE, Liu J, et al. A low-powered and highly selective trap for male *Aedes* (Diptera: Culicidae) surveillance: the Male Aedes Sound Trap. *J Med Entomol*. Jan 12 2021;58(1):408–415. <https://doi.org/10.1093/jme/tjaa151>.
 46. Staunton KM, Leiva D, Cruz A, et al. Outcomes from international field trials with Male Aedes Sound Traps: Frequency-dependent effectiveness in capturing target species in relation to bycatch abundance. *PLoS Neglected Trop Dis*. Feb 2021;15(2), e0009061. <https://doi.org/10.1371/journal.pntd.0009061>.
 47. Kweka EJ, Mwangonde BJ, Kimaro E, Msangi S, Massenga CP, Mahande AM. A resting box for outdoor sampling of adult *Anopheles arabiensis* in rice irrigation schemes of lower Moshi, northern Tanzania. *Malar J*. Apr 25 2009;8:82. <https://doi.org/10.1186/1475-2875-8-82>.
 48. Pombi M, Guelbeogo WM, Kreppel K, et al. The Sticky Resting Box, a new tool for studying resting behaviour of afrotropical malaria vectors. *Parasites Vectors*. May 29 2014;7:247. <https://doi.org/10.1186/1756-3305-7-247>.
 49. Fay RW, Donald AE. A preferred oviposition site as a surveillance method for *Aedes aegypti*. *Mosq news*. 1966;26(4):531–535.
 50. Mackay AJ, Amador M, Barrera R. An improved autocidal gravid ovitrap for the control and surveillance of *Aedes aegypti*. *Parasites Vectors*. Aug 6 2013;6(1):225. <https://doi.org/10.1186/1756-3305-6-225>.
 51. Ritchie SA, Long S, Hart A, Webb CE, Russell RC. An adulticidal sticky ovitrap for sampling container-breeding mosquitoes. *J Am Mosq Control Assoc*. Sep 2003;19(3):235–242.
 52. Lin LF, Lu WC, Cai SW, et al. The design and efficacy observation of new mosquito-ovitrap for monitoring of vector of dengue fever. *Chin J Vector Biol Control*. 2005;16(1):26–28.
 53. Barrera R, Amador M, Acevedo V, Caban B, Felix G, Mackay AJ. Use of the CDC autocidal gravid ovitrap to control and prevent outbreaks of *Aedes aegypti* (Diptera: culicidae). *J Med Entomol*. Jan 2014;51(1):145–154. <https://doi.org/10.1603/me13096>.
 54. Goodwin A, Glancey M, Ford T, et al. Development of a low-cost imaging system for remote mosquito surveillance. *Biomed Opt Express*. May 1 2020;11(5):2560–2569. <https://doi.org/10.1364/BOE.382391>.
 55. Guo X, Liu S, Liu X, et al. An improved ovitrap-based surveillance framework: facilitating cost-efficient monitoring and efficacy assessment of integrated vector management strategies for dengue outbreak control. *Parasites Vectors*. Sep 24 2025;18(1):380. <https://doi.org/10.1186/s13071-025-07002-8>.
 56. Degener CM, Staunton KM, Bossin H, et al. Evaluation of the new modular biogens BG-Pro mosquito trap in comparison to CDC, EVS, BG-Sentinel, and BG-Mosquitare traps. *J Am Mosq Control Assoc*. Dec 1 2021;37(4):224–241. <https://doi.org/10.2987/21-7003>.
 57. Fay RW, Perry AS. Laboratory studies of ovipositional preferences of *Aedes aegypti*. *Mosq news*. 1965;25(3):276–281.
 58. Codeco CT, Lima AW, Araujo SC, et al. Surveillance of *Aedes aegypti*: comparison of house index with four alternative traps. *PLoS Neglected Trop Dis*. Feb 2015;9(2), e0003475. <https://doi.org/10.1371/journal.pntd.0003475>.
 59. Ritchie SA, Buhagiar TS, Townsend M, et al. Field validation of the gravid Aedes trap (GAT) for collection of *Aedes aegypti* (Diptera: culicidae). *J Med Entomol*. Jan 2014; 51(1):210–219. <https://doi.org/10.1603/me13105>.
 60. Batool K, Xiao J, Xu Y, et al. Dengovirus oil suspension significantly improves the efficacy and duration of larvicidal activity against *Aedes albopictus*. *Viruses*. Feb 25 2022;14(3), 475. <https://doi.org/10.3390/v14030475>.
 61. Liu T, Xie YG, Lin F, et al. A long-lasting biological larvicide against the dengue vector mosquito *Aedes albopictus*. *Pest Manag Sci*. Feb 2021;77(2):741–748. <https://doi.org/10.1002/ps.6069>.
 62. Sasmita HI, Neoh KB, Yusmalinar S, et al. Ovitrap surveillance of dengue vector mosquitoes in Bandung City, West Java Province, Indonesia. *PLoS Neglected Trop Dis*. Oct 2021;15(10), e0009896. <https://doi.org/10.1371/journal.pntd.0009896>.
 63. Guo X, Li L, Ren W, et al. Modelling the dynamic basic reproduction number of dengue based on MOI of *Aedes albopictus* derived from a multi-site field investigation in Guangzhou, a subtropical region. *Parasites Vectors*. Feb 21 2024;17(1):79. <https://doi.org/10.1186/s13071-024-06121-y>.
 64. Rauhoft L, Sulesco T, Martins Afonso SM, et al. Large-scale performance assessment of the BG-Counter 2 used with two different mosquito traps. *Parasites Vectors*. Jun 27 2024;17(1):273. <https://doi.org/10.1186/s13071-024-06338-x>.
 65. Day CA, Richards SL, Reiskind MH, Doyle MS, Byrd BD. Context-dependent accuracy of the BG-Counter remote mosquito surveillance device in North Carolina. *J Am Mosq Control Assoc*. Jun 1 2020;36(2):74–80. <https://doi.org/10.2987/19-6903.1>.
 66. Njaime F, Maspero RC, Leandro AS, Maciel-de-Freitas R. Automated classification of mixed populations of *Aedes aegypti* and *Culex quinquefasciatus* mosquitoes under field conditions. *Parasites Vectors*. Sep 19 2024;17(1):399. <https://doi.org/10.1186/s13071-024-06417-z>.
 67. Micocci M, Manica M, Bernardini I, et al. An easier life to come for mosquito researchers: field-testing across Italy supports VECTrack system for automatic counting, identification and absolute density estimation of *Aedes albopictus* and *Culex pipiens* adults. *Parasites Vectors*. Oct 2 2024;17(1):409. <https://doi.org/10.1186/s13071-024-06479-z>.
 68. Sun Y, Lin Y, Zhao G, Svanberg S. Identification of flying insects in the spatial, spectral, and time domains with focus on mosquito imaging. *Sensors (Basel)*. May 11 2021;21(10), 3329. <https://doi.org/10.3390/s21103329>.
 69. Liu WL, Wang Y, Chen YX, et al. An IoT-based smart mosquito trap system embedded with real-time mosquito image processing by neural networks for mosquito surveillance. *Front Bioeng Biotechnol*. 2023;11, 1100968. <https://doi.org/10.3389/fbioe.2023.1100968>.
 70. Gonzalez Perez MI, Faulhaber B, Williams M, et al. Automated age grading of female *Culex pipiens* by an optical sensor system coupled to a mosquito trap. *Parasites Vectors*. Dec 18 2024;17(1):510. <https://doi.org/10.1186/s13071-024-06606-w>.
 71. Gonzalez-Perez MI, Faulhaber B, Aranda C, et al. Field evaluation of an automated mosquito surveillance system which classifies *Aedes* and *Culex* mosquitoes by genus and sex. *Parasites Vectors*. Mar 1 2024;17(1):97. <https://doi.org/10.1186/s13071-024-06177-w>.
 72. van Klink R, August T, Bas Y, et al. Emerging technologies revolutionise insect ecology and monitoring. *Trends Ecol Evol*. Oct 2022;37(10):872–885. <https://doi.org/10.1016/j.tree.2022.06.001>.
 73. Newson SE, Bas Y, Murray A, Gillings S. Potential for coupling the monitoring of bush-crickets with established large-scale acoustic monitoring of bats. *Methods Ecol Evol*. 2017;8(9):1051–1062. <https://doi.org/10.1111/2041-210X.12720>.
 74. Rydhmer K, Bick E, Still L, et al. Automating insect monitoring using unsupervised near-infrared sensors. *Sci Rep*. Feb 16 2022;12(1):2603. <https://doi.org/10.1038/s41598-022-06439-6>.
 75. Stanton MC, Kalonde P, Zembere K, Hoek Spaans R, Jones CM. The application of drones for mosquito larval habitat identification in rural environments: a practical approach for malaria control? *Malar J*. May 31 2021;20(1):244. <https://doi.org/10.1186/s12936-021-03759-2>.
 76. Haddawy P, Wettayakorn P, Nonthaleerak B, et al. Large scale detailed mapping of dengue vector breeding sites using street view images. *PLoS Neglected Trop Dis*. Jul 2019;13(7), e0007555. <https://doi.org/10.1371/journal.pntd.0007555>.
 77. Carrasco-Escobar G, Manrique E, Ruiz-Cabrejos J, et al. High-accuracy detection of malaria vector larval habitats using drone-based multispectral imagery. *PLoS Neglected Trop Dis*. Jan 2019;13(1), e0007105. <https://doi.org/10.1371/journal.pntd.0007105>.
 78. Guo X, Luo L, Long Y, et al. Field investigation combined with modeling uncovers the ecological heterogeneity of *Aedes albopictus* habitats for strategically improving systematic management during urbanization. *Parasites Vectors*. Oct 25 2023;16(1):382. <https://doi.org/10.1186/s13071-023-05926-7>.
 79. Guo X, Zhao R, Yuan Z, et al. Design and application of insect-catching equipment based on unmanned aerial vehicle. *Chin J Vector Biol Control*. 2022;33(6):865–868.
 80. Bohmann K, Evans A, Gilbert MT, et al. Environmental DNA for wildlife biology and biodiversity monitoring. *Trends Ecol Evol*. Jun 2014;29(6):358–367. <https://doi.org/10.1016/j.tree.2014.04.003>.
 81. Liu M, Clarke LJ, Baker SC, Jordan GJ, Burrige CP. A practical guide to DNA metabarcoding for entomological ecologists. *Ecol Entomol*. 2020;45(3):373–385. <https://doi.org/10.1111/een.12831>.
 82. Batovska J, Lynch SE, Cogan NOI, et al. Effective mosquito and arbovirus surveillance using metabarcoding. *Mol Ecol Resour*. Jan 2018;18(1):32–40. <https://doi.org/10.1111/1755-0998.12682>.
 83. Gutierrez-Lopez R, Egeter B, Paupy C, et al. Monitoring mosquito richness in an understudied area: can environmental DNA metabarcoding be a complementary approach to adult trapping? *Bull Entomol Res*. Aug 2023;113(4):456–468. <https://doi.org/10.1017/S0007485323000147>.
 84. Schneider J, Valentini A, Dejean T, et al. Detection of invasive mosquito vectors using environmental DNA (eDNA) from water samples. *PLoS One*. 2016;11(9), e0162493. <https://doi.org/10.1371/journal.pone.0162493>.